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Sr.No. ~~85911~~

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)

MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:401 Course Section:1	Seat No: 2022600033

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Name of the candidate. Shreya Ketan Mehta

Candidate's UID number: 2 0 2 2 6 0 0 0 3 3

Examination class room number: 4 0 1

Invigilator's signature with date: 8/10/25

(To be filled in by the candidate)

EXAMINATION: BTECH MSE

(For example FY MCA/FY BTECH)

SEMESTER : 7

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE-AIML

DAY AND DATE: Wednesday, 8-10-25

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
3) Do not write anything in the column provided for the marks to be assigned by the examiners.
4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
12) The answer-books will be scrutinized before they are sent to examiners.
13) All answer-books supplied shall be returned whether written or blank.
14) Nothing shall be written on the question-paper.
15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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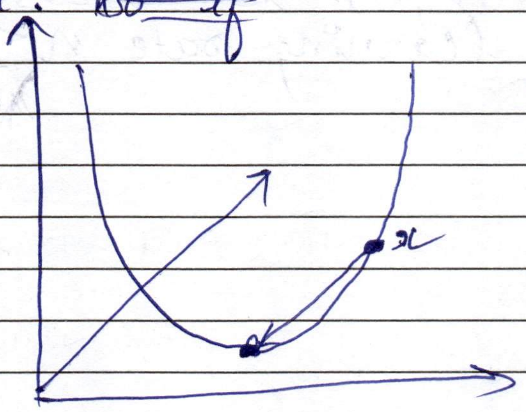
Q1 a) $f(w) = \frac{1}{2}\|xw - y\|^2 \rightarrow$ ~~of~~ loss function

$\therefore$ for the gradient update

$\therefore w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$

~~$\therefore \frac{\partial f}{\partial w} = \partial i$~~ For gradient descent, we try to lower the loss function in the direction of negative gradient. ~~So if~~



Every point x tries to reduce its loss to the minima $\rightarrow$ by updating its ~~$\therefore$ taking derivative of the loss function with~~ weights and ~~biad~~ bias.

$\therefore$ taking derivative of the loss function with respect to weight.

$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial a} \times \frac{\partial a}{\partial w}$ as weight ~~depepe~~ depends on the ~~activation function~~ o/p

$\therefore w_{new} = w_{old} - \eta\left(\frac{\partial L}{\partial a} \times \frac{\partial a}{\partial w}\right)$

$\frac{\partial L}{\partial a} = (xw - y)(+1)$, $\frac{\partial a}{\partial w}$

Similarly new bias can be updated as $b_{new} = b_{old} - \eta \frac{\partial L}{\partial b}$

loss depends on o/p $\rightarrow$ o/p depends on activation & that depends on bias.

$\therefore \frac{\partial L}{\partial b} = \frac{\partial L}{\partial O} \times \frac{\partial O}{\partial a} \times \frac{\partial a}{\partial b}$

$\therefore b_{new} = b_{old} - \eta\left(\frac{\partial L}{\partial O} \times \frac{\partial O}{\partial a} \times \frac{\partial a}{\partial b}\right)$

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learning rate $\eta < \frac{2}{\lambda_{max}}$

eigen value of $x^T x =$

$\lambda^2 -$ (sum of diagonal elements) + det of ses matrix) $= 0$

$\therefore$ $\cancel{\eta < \frac{2}{\lambda_{max}}}$ $\because$ for getting maximum convergence, it is necessary that the learning rate $\eta < \frac{2}{\lambda_{max}}$.

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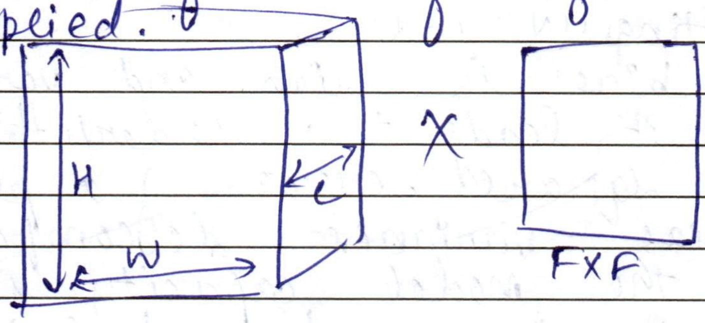
Q1b) The bias-variance decomposition is that if the bias is low and the variance is high then it leads to Overfitting
If the bias is high and variance is low, it leads to underfitting.
∴ ~~For squared-error of a predictor~~, the ~~bias~~ variance decomposition is:
when the model capacity is too high i.e. it learns the training data very well because of dence layers the model gives a good accuracy for training data, but when it is fed the test data, it performs poorly.
Thus it negatively affects generalization As there is overfitting. and cannot accurately tell weather the model is good because of test data.

Q2a) For a standard 2D convolution, input size given is $H \times W \times C$ and K filters of size $F \times F$ are applied.

∴

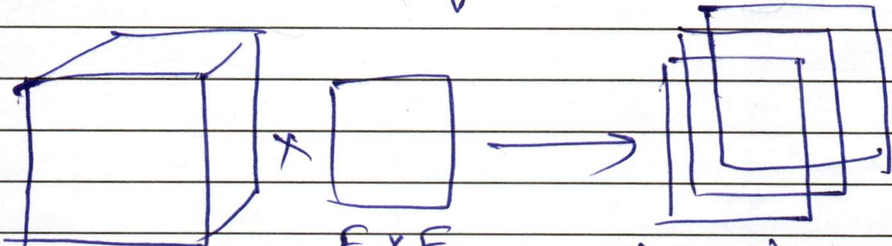


We will convol the filter over the input with the given stride and calculate ~~based on~~ each pixel of the feature map

∴ $H' = H - F + 1$

$W' = W - F + 1$

This is applied for K times so the final feature map will be ~~$h-f+1$~~ $(H-F+1) \times (W-F+1) \times K$



~~In a depthwise seperable convolutions, even the depth is considered while moving the filter over input.~~

Computational complexity of 2D convolution
$= K * (H-F+1) \times (W-F+1)$ as stride is 1.

Depthwise-seperable requires more computation as we consider depth C as well while applying filters

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depthwise separable convolutions are critical in modern architectures because they filter the features more accurately hence better for tasks which require high perfection.

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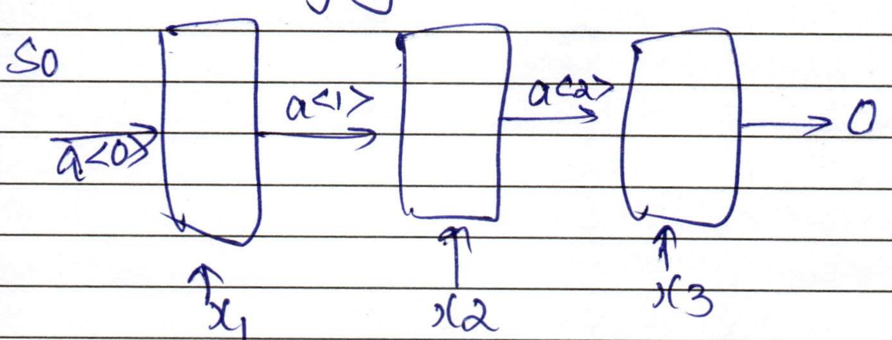
Q3a) For an RNN, recursive gradient expression $\frac{\partial L}{\partial h_{t-k}}$.

$$a = \sigma(W_{aa} h_{t-1} + W_{ax} x + b)$$

$$\cancel{y} = \frac{\partial L}{\partial h_{t-k}} = \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_2} \times \frac{\partial a_2}{\partial a_1} \times \frac{\partial a_1}{\partial h_{t-k}}$$

In LSTMs, the forget gate decides what to forget and what to keep in the hidden state, so RNNs cannot remember long-distance words,
as the gradients decrease over time, so the change in weights become negligible



here by the time it reaches last layer the gradient has diminished significantly so the loss doesn't decrease much.
LSTM remembers the words for longer distance ~~and hence cannot take long time~~

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Q3b) For GRU,

update gate $z_t = \sigma(W_z[h_{t-1}, x] + b_z)$

reset gate $r_t = \sigma(W_r[h_{t-1}, x] + b_r)$

Candidate state $= \tanh(W_c[r_t \cdot h_{t-1}, x] + b_c)$

$\therefore$ ~~$h_t = (1-z_t) h_{t-1} + (z_t) \cdot \tilde{C}_t$~~

~~update rule = $(1-z_t) h_{t-1} + z_t \tilde{C}_t$~~

$h_t = \sigma(W_t[h_{t-1}, x] + b_t)$

update rule $=$ ~~$(1-z_t)\tilde{C}_t + z_t h_t$~~

$= (1-z_t) h_t + z_t \tilde{C}_t$

The update rule of a GRU is a convex combination of the previous hidden state or the candidate state, i.e if the update is near 0, that means no update and if it is near to 1 then update with the new candidate state.

LSTMs have 3 gates and 2 states

forget gate $f_t = \sigma(W_f[h_{t-1}, x] + b_f)$

input gate $i_t = \sigma(W_i[h_{t-1}, x] + b_f)$

output gate $O_t = \sigma(W_o[h_{t-1}, x] + b_o)$

Candidate state $\tilde{C}_t = \tanh(W_c[h_{t-1}, x] + b_c)$

$\therefore$ cell state $= f_t \times C_{t-1} + i_t \times \tilde{C}_t$

hidden state $= \sigma(W_h[h_{t-1}, x] + b_h)$

~~out output = softmax(hidden state + cellstate)~~

Thus since it has 3 gates and 2 states cell state and hidden state ~~LSTMs have~~ GRUs have fewer parameters compared to LSTMs.

It ~~is~~ proves generalization like for example Tom is happy. He got good marks, In this to remember who He belongs to, it is necessary to ~~a~~ remember the input for a long time & this is achieved by GRUs with less no. of gates thus it is computationally better.

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